Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam D · Form 1: Multi-step reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. Two cars moving at speed u see an obstacle. A brakes at magnitude b immediately. B waits τ and then brakes at 2b. When does B stop in a shorter distance?

(A) τ<u/(4b)

(B) τ>u/b

(C) τ<u/(2b)

(D) always

2. From a balcony 4.9 m above ground, a ball is thrown vertically upward at 19.6 m/s. Ignore drag; ground is y = 0. Find the highest position above ground. Use g = 9.8 m/s².

(A) 12.25 m

(B) 24.5 m

(C) -24.5 m

(D) 49 m

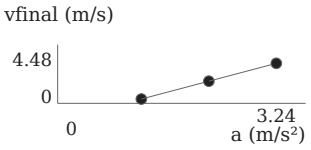
3. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1.24, 0.48); (2.24, 2.48); (3.24, 4.48). A logger displays twice each true horizontal reading. What slope appears on the displayed axes?

(A) 1

(B) 0.5

(C) 2

(D) 1.5



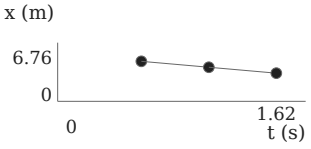
4. Photogates record the position of a cart traveling at constant velocity. Reverse the positive position direction only. Under the revised model, what is Y at X=1.12?

(A) -5.34

(B) -6.6

(C) -4.5

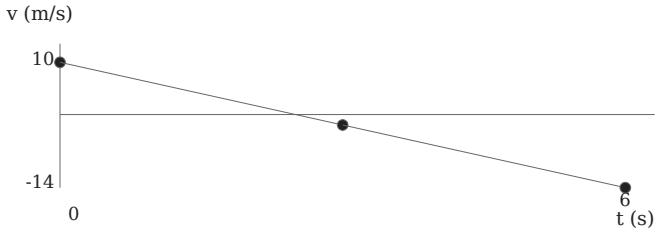
(D) -5.76



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records a cart with x0 = -5 m and v0 = +10 m/s. It has constant a = -4 m/s² for 6 s. Right is positive.

a. Describe the shapes and initial values of x–t, v–t, and a–t. **(1 pt)**



b. Find final position, displacement, and distance; explain any reversal. **(3 pt)**

6. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.62, 7.14); (1.12, 5.64); (1.62, 4.14).

a. Build the model for runner separation from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the runners' velocities but leave starting positions unchanged. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A cart moves right at speed u and experiences constant acceleration -b, where b > 0. Derive stopping time and distance, then compare them when u doubles. **(3 pt)**